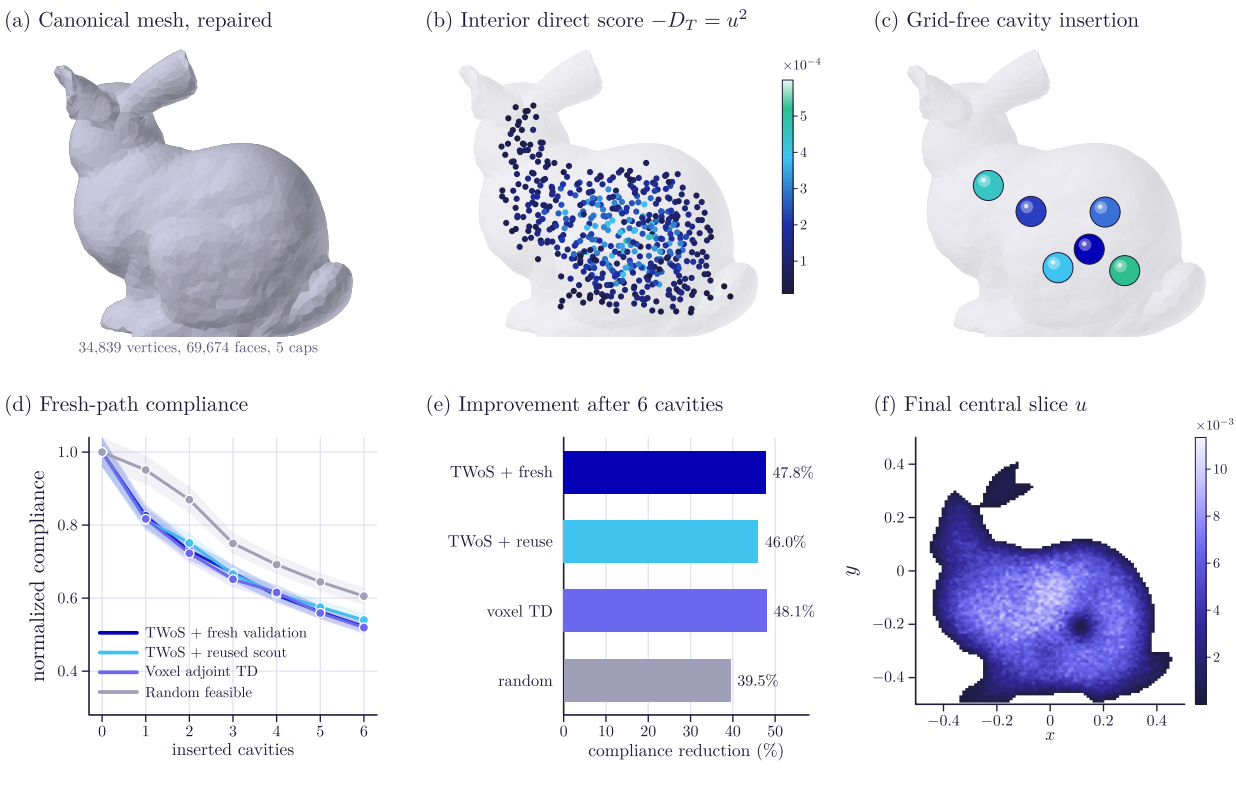

(a) Canonical mesh, repaired
34,839 vertices, 69,674 faces, 5 caps
(b) Interior direct score $-D_T = u^2$
×10⁻⁴
5
4
3
2
1
(c) Grid-free cavity insertion
(d) Fresh-path compliance
normalized compliance
1.0
0.8
0.6
0.4
0
1
2
3
4
5
6
inserted cavities
TWoS + fresh validation
TWoS + reused scout
Voxel adjoint TD
Random feasible
(e) Improvement after 6 cavities
TWoS + fresh
47.8%
TWoS + reuse
46.0%
voxel TD
48.1%
random
39.5%
0
10
20
30
40
50
compliance reduction (%)
(f) Final central slice u
×10⁻³
10
8
6
4
2
0.4
0.2
0
−0.2
−0.4
y
x